2D Pose Graph

Problem formulation

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

N number of poses

z_{ij} is the virtual measurement between node i & j

$\therefore z$ is the vector of z_{ij}

$\hat{z}_{ij}(x_i, x_j)$ is the prediction of z_{ij}, it is usually a relative transformation between the nodes.

** we need to find $p(x|z,u)$

$$p(x|z,u) = p(x|z) \quad [\text{odometry measurements are taken into } z]$$

$$p(x|z) = \frac{p(z|x)\,p(x)}{p(z)} \quad [\text{Bayes' Rule}]$$

$$\propto p(z|x)\,p(x)$$

likelihood, prior

$$\propto p(z|x) \quad [\text{assume no prior knowledge}]$$

** We need to find x^* which maximizes $p(x|z,u)$

$$x^* = \arg\max_x \left(p(z|x)\right)$$

We have a model for z:

$$z = \hat{z} + v, \quad v \sim \mathcal{N}(0, Q)$$

$$\therefore p(z|x) \sim \mathcal{N}(\hat{z}, Q)$$

$$\Rightarrow p(z_{ij}|x_i, x_j) \sim \mathcal{N}(\hat{z}_{ij}(x_i, x_j), Q_{ij})$$

$$\Rightarrow -\ln\left(p(z_{ij}|x_i, x_j)\right) = \frac{1}{2}\left[(z_{ij} - \hat{z}_{ij})^T Q^{-1} (z_{ij} - \hat{z}_{ij})\right]$$

$$\therefore x^* = \arg\min_x \left(\underbrace{(z_{ij} - \hat{z}_{ij})^T}_{e_{ij}^T} Q^{-1} \underbrace{(z_{ij} - \hat{z}_{ij})}_{e_{ij}} \right)$$

$$\Rightarrow x^* = \underset{x}{\operatorname{argmin}} \left(\sum_{ij} e_{ij}^T \Omega^{-1} e_{ij} \right)$$

$$\Rightarrow x^* = \underset{x}{\operatorname{argmin}} \left(F(x) \right)$$

↖ objective functn.

$$\Rightarrow x^* = \underset{x}{\operatorname{argmin}} \left(\sum_{(i,j)} e_{ij}^T \Omega_{ij}^{-1} e_{ij} \right)$$

$$\Rightarrow x^* = \underset{x}{\operatorname{argmin}} \left(F(x) \right)$$ ← objective functn.

the x in $F(x)$ can be written as some initial guess + addtn;

$$x = \breve{x} + \Delta x$$

$$F(x) = F(\breve{x} + \Delta x)$$

for a pair i,j;

$$F_{ij}(x) = F_{ij}(\breve{x} + \Delta x)$$

$$= e_{ij}^T(\breve{x}_i + \Delta x_i, \breve{x}_j + \Delta x_j)\, \Omega_{ij}^{-1}\, e_{ij}(\breve{x}_i + \Delta x_i, \breve{x}_j + \Delta x_j)$$

$$= e_{ij}^T(\breve{x} + \Delta x)\, \Omega_{ij}^{-1}\, e_{ij}(\breve{x} + \Delta x)$$

Taylor expanding the error,

$$= (e_{ij} + J_{ij}\Delta x)^T \Omega_{ij}^{-1} (e_{ij} + J_{ij}\Delta x)$$

$$= \underbrace{e_{ij}^T \Omega_{ij}^{-1} e_{ij}}_{c \text{ term}} + \underbrace{2 e_{ij}^T \Omega_{ij}^{-1} J_{ij}}_{\substack{\text{coeff of} \\ 1^{st} \text{ order term} \\ (b_{ij})}} \Delta x + \underbrace{\Delta x^T J_{ij}^T \Omega^{-1} J_{ij}}_{\substack{\text{coeff of} \\ \text{quadratic} \\ \text{term.} \\ (H_{ij})}} \Delta x$$

$$= c_{ij} + 2 b_{ij} \Delta x + \Delta x^T H_{ij} \Delta x$$

$$F(x) = \sum_{(i,j)} c_{ij} + 2 b_{ij} \Delta x + \Delta x^T H_{ij} \Delta x$$

$$\boxed{= C + 2b\Delta x + \Delta x^T H \Delta x}$$

to solve for Δx, we take the derivative at $= 0$ (extremum for quadratic)

$$\boxed{H \Delta x^* = -b} \longrightarrow \text{Least squares format}$$

After finding Δx^*, we add with the guess until it is good enough $[x^* = \breve{x} + \Delta x]$

On a Manifold

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} \qquad X_i \in SE(2) \text{ or } SE(3)$$

Z_{ij} is the measurement between node i & j

Z is the set $\{Z_{ij}\}$

$\hat{Z}_{ij}$ is the prediction of Z_{ij}, typically relative transformation between node i & j

Find X that best satisfies (maximizes) $P(X \mid Z, U)$.

$$P(X \mid Z, U) \propto P(X \mid Z)$$
$$\propto P(Z \mid X) P(X) \text{ [Bayes rule and then uniformity]}$$
$$\propto P(Z \mid X) \text{ [Assume no given prior]}$$

$$X^* = \arg\max_X \left(P(Z \mid X)\right)$$

$Z = \hat{Z}\, Exp(v)$ where $v \sim \mathcal{N}(0, Q)$ & $Exp(x)$ takes a ~~vector~~ Lie algebra element and returns its corresponding Lie group element.

$$Z_{ij} = \hat{Z}_{ij}(X_i, X_j)\, Exp(v)$$
$$Z_{ij} = X_i^{-1} X_j\, Exp(v)$$

$$P(Z \mid X) \sim \mathcal{N}(\hat{Z}, Q)$$
$$= \prod_{(i,j)} p(z_{ij} \mid x_i, x_j)$$
$$-\ln P(Z \mid X) = -\frac{1}{2} \Big(\underbrace{\left(\mathrm{Log}(\hat{Z}_{ij}^{-1} Z_{ij})\right)^T}_{e_{ij}^T} \underbrace{Q_{ij}^{-1}}_{\Omega_{ij}} \underbrace{\left(\mathrm{Log}(\hat{Z}_{ij}^{-1} Z_{ij})\right)}_{e_{ij}} \Big) + \text{Const}$$

$$X^* = \arg\min_X \left(\sum_{(i,j)} e_{ij}^T \Omega_{ij} e_{ij} \right)$$
$$= \arg\min_X \left(e^T \Omega e \right)$$
$$= \arg\min \left(F(x) \right)$$

Some Lie theory Conventions.

$X \in G$ $\quad$ $Y \in G$

$x^\wedge \in \mathfrak{g}$ $\quad$ $y^\wedge \in \mathfrak{g}$

$x \in \mathbb{R}^d$ $\quad$ $y \in \mathbb{R}^d$

$X \oplus Y = X \operatorname{Exp}(Y) = X \exp(y^\wedge) \in G$

$Y \ominus X = \operatorname{Log}(X^{-1} \cdot Y) = \log(X^{-1} \cdot Y)^\vee \in \mathfrak{g} \cong \mathbb{R}^d$

The goal is to find $\frac{\partial e_{ij}}{\partial \delta x_i}$ & $\frac{\partial e_{ij}}{\partial \delta x_j}$

we define the residual, e_{ij}, as

$$e_{ij} = \log(Z_{ij}^{-1} X_i^{-1} X_j)^\vee \in \mathbb{R}^6$$

$$\Delta x = \begin{pmatrix} \delta x_1 \\ \vdots \\ \delta x_n \end{pmatrix}$$



Now we first introduce a small perturbation,

$$X_i \to X_i \exp(\delta \hat{x_i}) \qquad X_j \to X_j \exp(\delta \hat{x_j})$$

→ $f(x_0+\delta) \approx f(x_0) + f'(x_0)\delta$

The residual now becomes a function of the perturbation,

$$e_{ij} \to e_{ij}(\Delta x) \to e_{ij}(\delta x_i, \delta x_j) \approx e_{ij} + J_{ij} \Delta x \approx e_{ij} + A_{ij}\delta x_i + B_{ij}\delta x_j$$

$\to e_{ij}(X \boxplus \Delta x)$

We need to find A_{ij} & B_{ij} and by extension J_{ij}.

$$e_{ij}(\delta x_i, \delta x_j) = \log(Z_{ij}^{-1}(X_i \exp(\delta \hat{x_i}))^{-1} X_j \exp(\delta \hat{x_j}))^\vee$$

$$= \log(Z_{ij}^{-1} \exp(\delta \hat{x_i})^{-1} X_i^{-1} X_j \exp(\delta \hat{x_j}))^\vee$$

$$= \log(Z_{ij}^{-1} \exp(-\delta \hat{x_i}) X_i^{-1} X_j \exp(\delta \hat{x_j}))^\vee$$

Let,

$T = X_i^{-1} X_j$, $E = Z_{ij}^{-1} T = Z_{ij}^{-1} T = Z_{ij}^{-1} X_i^{-1} X_j$, $e_{ij} = \log(E)^\vee$

$T^{-1} = X_j^{-1} X_i$ $\qquad E = \exp(e_{ij}\hat{})$

$$e_{ij}(\delta x_i, \delta x_j) = \log(Z_{ij}^{-1} \underline{\exp(-\delta \hat{x_i})} T \exp(\delta \hat{x}))^\vee$$

↳ we have to move this.

from the residual,

$$\exp(-\delta x_i^\wedge) T = (T T^{-1}) \exp(-\delta x_i^\wedge) T = T \left(T^{-1} \exp(-\delta x_i^\wedge) T \right)$$

$$= T \exp\left((-Ad_{T^{-1}} \delta x_i)^\wedge \right)$$

$$\therefore e_{ij} = \log \left(Z_{ij}^{-1} T \exp\left((-Ad_{T^{-1}} \delta x_i)^\wedge \right) \exp(\delta x_j^\wedge) \right)^\vee$$

$$= \log \left(E \exp\left((-Ad_{T^{-1}} \delta x_i)^\wedge \right) \exp(\delta x_j^\wedge) \right)^\vee$$

$X \in SE(3)$
$a \in se(3)$

$X \exp(a) = \exp((Ad_X a)^\wedge) X$

overall perturbation made by local transform expressed in world perturbation

$\exp((Ad_X a)^\wedge) = X \exp(a) X^{-1}$

$e_{ij} = (\log E)^\vee$
$E = \exp(e_{ij}^\wedge)$

Now to expand each part individually,

setting $\delta x_j = 0$, $\exp(0) = I$

$$e_{ij}(\delta x_i, 0) = \log \left(E \exp\left((-Ad_{T^{-1}} \delta x_i)^\wedge \right) \right)^\vee$$

$$= \log \left(\exp(e_{ij}^\wedge) \exp\left((-Ad_{T^{-1}} \delta x_i)^\wedge \right) \right)^\vee$$

Applying the BCH approximation,

$$\approx e_{ij} + J_r^{-1}(e_{ij}) \left(-Ad_{T^{-1}} \delta x_i \right)$$

$$= e_{ij} + \underbrace{\left(-J_r^{-1}(e_{ij}) Ad_{T^{-1}} \right)}_{A_{ij}} \delta x_i \rightarrow \text{similar to the form } e_{ij} + J \Delta x$$

Now setting $\delta x_i = 0$,

$$e_{ij}(0, \delta x_j) = \log \left(\exp(e_{ij}^\wedge) \exp(\delta x_j^\wedge) \right)^\vee$$

Applying BCH approximation,

$$\approx e_{ij} + \underbrace{J_r^{-1}(e_{ij})}_{B_{ij}} \delta x_j \rightarrow \text{similar to form } e_{ij} + J \Delta x$$

$$e_{ij}(\delta x_i, \delta x_j) = e_{ij} + 0 + 0 \cdots + \underbrace{\left(-J_r^{-1}(e_{ij}) Ad_{T^{-1}} \right)}_{A_{ij}} \delta x_i + 0 + 0 + 0 + \cdots \underbrace{J_r^{-1}(e_{ij})}_{B_{ij}} \delta x_j + 0 + 0 -$$

$$= e_{ij} + J_{ij} \Delta x$$

where $\Delta x = (\delta x_1 \cdots \delta x_i \sim \delta x_j \sim \delta x_n)^T$ & $J = (0 \cdots A_{ij} \cdots B_{ij} \cdots 0)$